

Normalization

PK=(user_id, event_id, allocation_id, request_id)

1. 1NF TABLE :

Column	Notes
user_id	PK (part 1 of 4)
event_id	PK (part 2 of 4)
allocation_id	PK (part 3 of 4)
request_id	PK (part 4 of 4)
full_name, email, password, role	User details
created_at, clerk_id	User meta
society_id, society_name, description	Society details
society_head_id, established_date	Society meta
event_name, event_description, event_date	Event details
event_status	Event status
venue_id, venue_name	Venue details
resource_id, item_name, total_inventory	Resource details
quantity_requested, allocation_status	Allocation details
requested_role, request_status	Request details
reviewed_by, request_created_at, reviewed_at	Request meta
student_id, registration_date	Registration details

Anomalies : Even in 1NF, the table has severe partial dependencies: most columns depend on only part of the composite PK. For example, full_name depends only on user_id, not on event_id, allocation_id, or request_id. This violates 2NF.

2. 2NF CHECKING FOR PARTIAL DEPENDENCIES

Attributes	Depends Only On	Action
full_name, email, password, role, created_at, clerk_id	user_id	Extract → USERS
society_name, description, society_head_id, established_date	society_id	Extract → SOCIETIES
event_name, event_description, event_date, event_status, venue_id	event_id	Extract → EVENTS
venue_name	venue_id	Extract → VENUES
item_name, total_inventory	resource_id	Extract → RESOURCES
quantity_requested, allocation_status event_id (FK), resource_id (FK)	allocation_id	Extract → RESOURCE_ALLOCATION
requested_role, request_status, reviewed_by, request_created_at, reviewed_at, user_id (FK)	request_id	Extract → REQUESTS
registration_date	(user_id, event_id)	Extract → EVENT_REGISTRATIONS

RESULTING 2NF TABLES

Users :

Column	Data Type	Notes
<u>user_id</u>	UUID	Primary Key
full_name	TEXT	NOT NULL
email	TEXT	UNIQUE, NOT NULL
password	TEXT	
role	TEXT	Default: 'Student'
created_at	TIMESTAMP	
clerk_id	TEXT	UNIQUE

Societies :

Column	Data Type	Notes
<u>society_id</u>	SERIAL	Primary Key
society_name	TEXT	UNIQUE, NOT NULL
description	TEXT	
society_head_id	UUID	FK → users(user_id)
established_date	DATE	

Venues :

Column	Data Type	Notes
<u>venue_id</u>	INTEGER	Primary Key
venue_name	TEXT	

Events:

Column	Data Type	Notes
<u>event_id</u>	SERIAL	Primary Key
event_name	TEXT	NOT NULL
event_description	TEXT	
event_date	DATE	NOT NULL
society_id	INTEGER	FK → societies(society_id)
venue_id	INTEGER	FK → venues(venue_id)
status	TEXT	Default: 'Scheduled'

Resources:

Column	Data Type	Notes
<u>resource_id</u>	SERIAL	Primary Key
item_name	TEXT	NOT NULL
total_inventory	INTEGER	NOT NULL, ≥ 0

Resource Allocations:

Column	Data Type	Notes
<u>allocation_id</u>	SERIAL	Primary Key
<i>event_id</i>	INTEGER	FK → events(event_id)
<i>resource_id</i>	INTEGER	FK → resources(resource_id)
quantity_requested	INTEGER	NOT NULL, > 0
allocation_status	TEXT	Default: 'Pending'

Requests:

Column	Data Type	Notes
<u>request_id</u>	SERIAL	Primary Key
<i>user_id</i>	UUID	FK → users(user_id)
requested_role	TEXT	NOT NULL
status	TEXT	Default: 'Pending'
<i>reviewed_by</i>	UUID	FK → users(user_id)
created_at	TIMESTAMP	
reviewed_at	TIMESTAMP	

Event Registration:

Column	Data Type	Notes
<u>event_id</u>	INTEGER	PK (part 1), FK → events(event_id)
<u>student_id</u>	TEXT	PK (part 2)
registration_date	TIMESTAMP	Default: CURRENT_TIMESTAMP

Anomalies: After reaching 2NF, partial dependencies are eliminated. However, transitive dependencies may still exist — a non-key attribute could depend on another non-key attribute rather than directly on the primary key. This violates 3NF.

3. 3NF CHECKING FOR TRANSITIVE DEPENDENCIES

Table	Potential Transitive Dep.	Verdict
USERS	email → nothing non-key; role is standalone	OK
SOCIETIES	society_head_id is a FK (UUID), not a derived value	OK
VENUES	Only one non-key: venue_name	OK
EVENTS	All non-keys (event_name, event_date, status, society_id, venue_id) depend directly on event_id	OK
RESOURCES	item_name, total_inventory depend directly on resource_id	OK
RESOURCE_ALLOC.	quantity_requested, allocation_status de- pend on allocation_id	OK
REQUESTS	All non-keys depend on request_id directly	OK
EVENT_REGISTRATIONS	registration_date depends on (event_id, student_id)	OK

No transitive dependencies were found. The 2NF tables are already in 3NF because:

- Each table was cleanly decomposed around a single real-world entity during the 2NF step.
- Foreign keys (like venue_id in Events) are part of a relationship, not a transitive dependency — they do not cause non-key attributes to depend on other non-key attributes.
- All non-key attributes describe the entity whose PK governs the table, and nothing else. The 2NF tables carry forward to 3NF unchanged.

4. BCNF CHECKING FOR BCNF VIOLATIONS

Table	Functional Dependencies to Check	Verdict
USERS	$\text{email} \rightarrow \text{row}$ (email is a candidate key / UNIQUE) $\text{clerk_id} \rightarrow \text{row}$ (also UNIQUE)	OK
SOCIETIES	$\text{society_name} \rightarrow \text{row}$ (UNIQUE constraint = candidate key) $\text{society_head_id} \rightarrow \text{row}$ (UNIQUE = one head per society)	OK
VENUES	$\text{venue_id} \rightarrow \text{venue_name}$: trivial, PK determines	OK
EVENTS	All non-key attributes determined by event_id (PK) only	OK
RESOURCES	$\text{resource_id} \rightarrow \text{all}$; no other determinants	OK
RESOURCE_ALLOC.	$\text{allocation_id} \rightarrow \text{all}$; ($\text{event_id}, \text{resource_id}$) is also a candidate key, but it determines the same row	OK
REQUESTS	$\text{request_id} \rightarrow \text{all}$; no non-key determines another attribute	OK
EVENT_REGISTRATIONS	($\text{event_id}, \text{student_id}$) is PK; only one non-key: registration_date	OK

All tables satisfy BCNF. Key observations:

- In Users, email and clerk_id are declared UNIQUE, making them candidate keys —every determinant is therefore a superkey.
- In Societies, society_name (UNIQUE) and society_head_id (UNIQUE) are candidate keys. They determine the row, which is allowed in BCNF.
- In Resource_Allocations, ($\text{event_id}, \text{resource_id}$) is an alternate candidate key for the same row already identified by allocation_id. Since both are superkeys, no violation exists.

- No table contains a non-superkey attribute set that determines any other attribute. The 3NF schema is already in BCNF — no further decomposition is needed. This is the final normalized schema.

Table	Primary Key	Key Relationships
USERS	user_id (UUID)	Referenced by SOCIETIES, REQUESTS
SOCIETIES	society_id (SERIAL)	→ USERS (head); ← EVENTS
VENUES	venue_id (INTEGER)	← EVENTS
EVENTS	event_id (SERIAL)	→ SOCIETIES, VENUES
RESOURCES	resource_id (SERIAL)	← RESOURCE_ALLOCATIONS
RESOURCE_ALLOCATIONS	allocation_id (SERIAL)	→ EVENTS, RESOURCES
REQUESTS	request_id (SERIAL)	→ USERS (requester & reviewer)
EVENT_REGISTRATIONS	(event_id, student_id)	→ EVENTS